4.3.5 Use of amount of substance in relation to volumes of gases (chemistry only) (HT only)

Content	Key opportunities for skills development
Equal amounts in moles of gases occupy the same volume under the same conditions of temperature and pressure. The volume of one mole of any gas at room temperature and pressure (20^{o}C and 1 atmosphere pressure) is 24 dm^3. The volumes of gaseous reactants and products can be calculated from the balanced equation for the reaction. Students should be able to: • calculate the volume of a gas at room temperature and pressure from its mass and relative formula mass • calculate volumes of gaseous reactants and products from a balanced equation and a given volume of a gaseous reactant or product • change the subject of a mathematical equation.	WS 1.2, 4.1, 4.2, 4.3, 4.6 MS 1a Recognise and use expressions in decimal form. MS 1c Use ratios, fractions and percentages. MS 3b Change the subject of an equation. MS 3c Substitute numerical values into algebraic equations using appropriate units for physical quantities.